

Correctness summary: current reference policy

evidence family	max relative L2	max absolute error	reference
symbolic/direct evaluation	1.618e-16	0.007812	C++ cpu_serial matrix
backend tradeoff sweep	1.517e-16	0.007812	C++ cpu_serial matrix
external Abaqus/MATLAB matrix	not used	not used	future cross-check

All current correctness rows compare candidate matrices against the C++ cpu_serial assembled matrix.